AERO60492 – Autonomous Mobile Robots

Lecture 5 – Bandwidth and Rate

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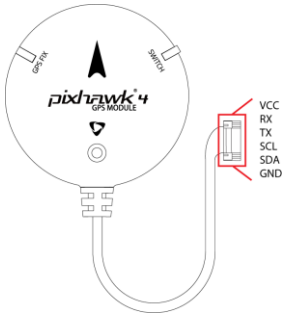
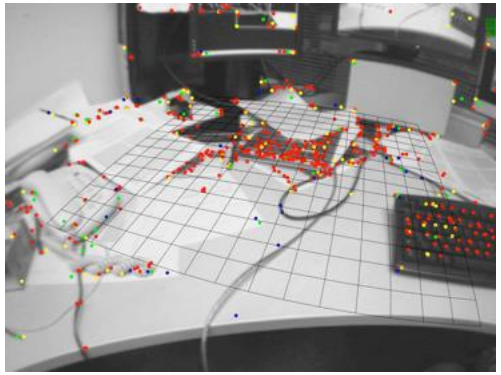
Aims

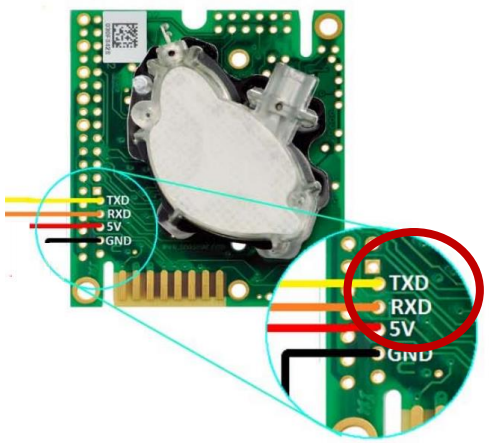
- Understand terminology
- Understand effect of communication overheads
- Design appropriate sampling rates
- Problems with too low rate



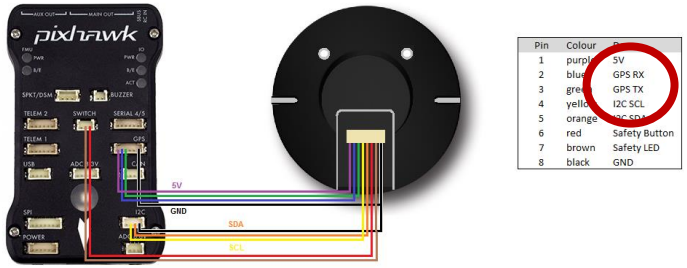


Bit rate

- The lowest level of communication
- The number of 1s and 0s transmitted per second
- Handled by dedicated hardware
- Settings must be correct
- One example is UART
  - Seen on many robotics sensors
  - Indicated by Rx,Tx markings



K30 CO2 concentration sensor



Here+ GPS connection to autopilot

Waveshare laser range finder

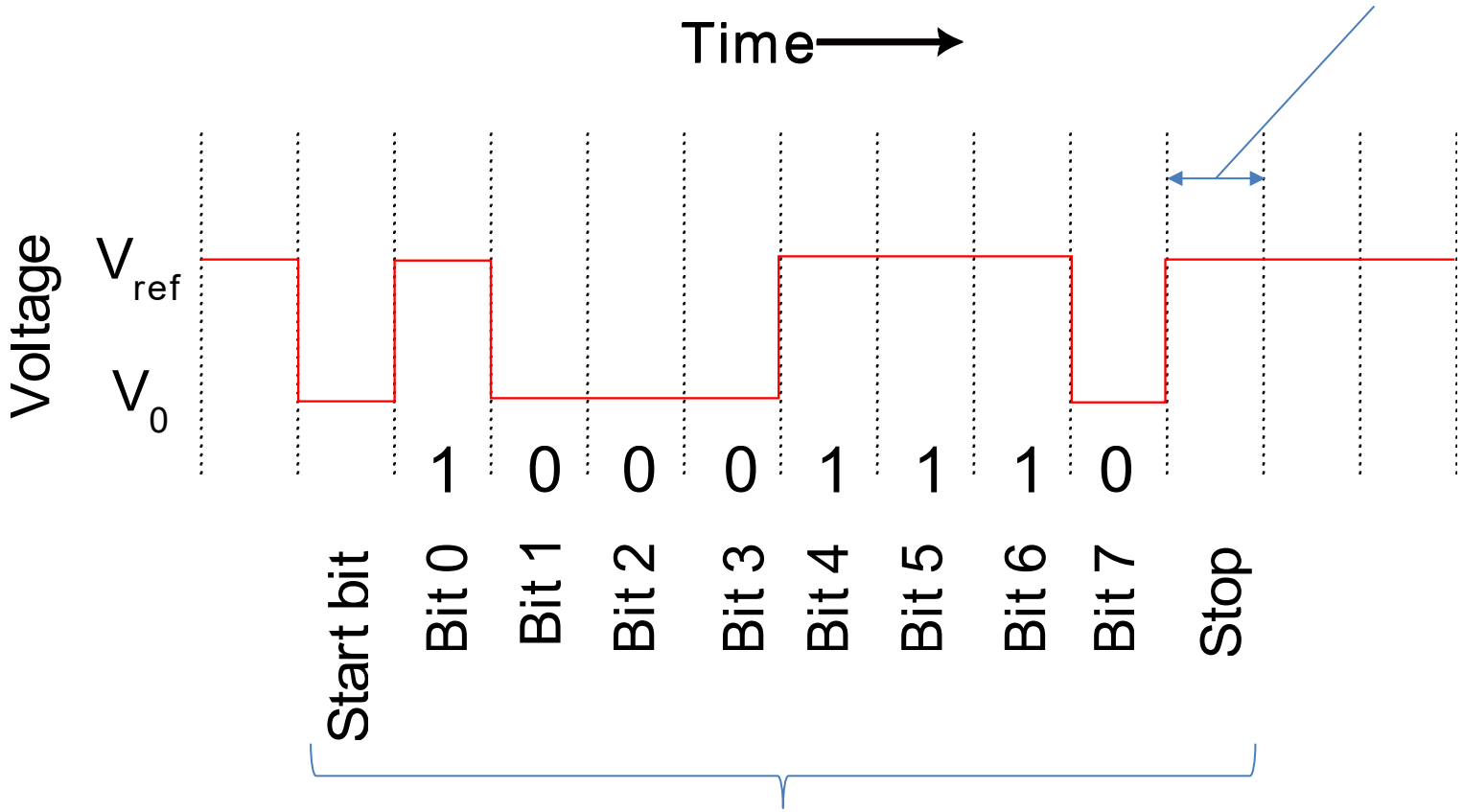
MANCHESTER 1824
The University of Manchester

UART

- Universal Asynchronous Receiver/Transmitter
  - A simple form of serial communication
- ‘Asynchronous’ means there is no clock signal shared
  - Both ends must agree on structure and timings
- Several different structures but all follow the same pattern
- Defined with:
  - Baud rate
  - Number of bits per frame
  - Parity bit (N-none, E-even, O-odd)
  - Number of stop bits

UART e.g. 9600-8N1

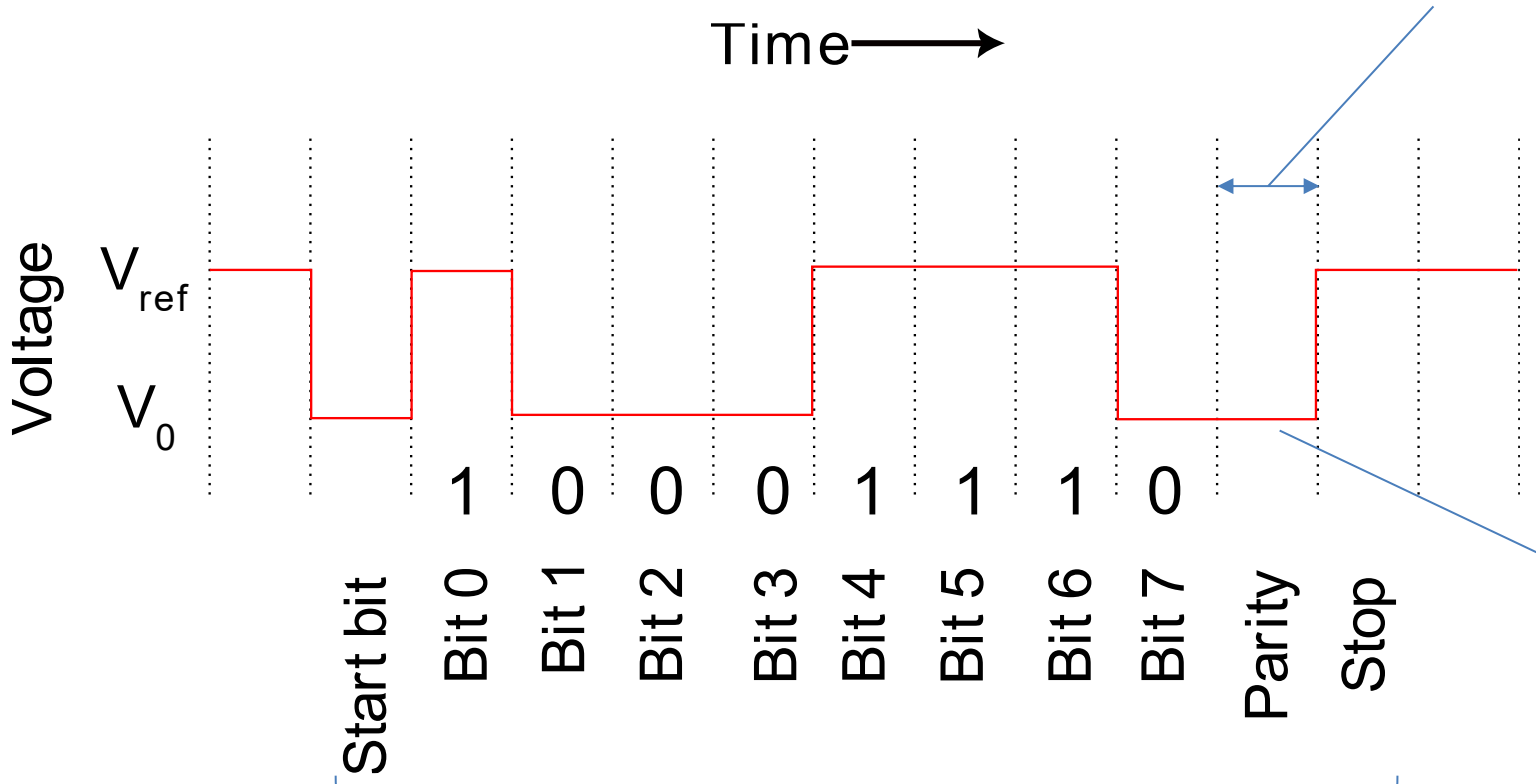
$\frac{1}{baudrate} \approx 104\mu s$ for 9600



Total bits per frame = 10
Useful bit per frame = 8
80% utilisation, 20% overhead

UART e.g. 9600-8E1

$$\frac{1}{baudrate} \approx 104\mu s$$



Parity bit set high or low to ensure an Even number of 1s in data bits+parity (i.e. not including start/stop bits)

'overheads' are the parts of the bits/bytes which are there to facilitate reliable transmissions and not the encoded data values

Total bits per frame = 11
Useful bit per frame = 8
73% utilisation, 27% overhead

What is the byte rate?

Baud = (number of bytes) x (total bits per frame) x (output rate Hz)

$$\therefore \quad \frac{\text{Baud}}{\text{(total bits per frame)}} = \text{(number of bytes) x (output rate Hz)}$$

∴ Available data budget a balance of data rate or data packet size

Transmission speed			
Bauds	**Bit/s**	**Bit duration**	**Speed**
50 Bd	50 bits/s	20.000 ms	6.25 bytes/s
75 Bd	75 bits/s	13.333 ms	9.375 bytes/s
110 Bd	110 bits/s	9.091 ms	13.75 bytes/s
134 Bd	134 bits/s	7.463 ms	16.75 bytes/s
150 Bd	150 bits/s	6.667 ms	18.75 bytes/s
200 Bd	200 bits/s	5.000 ms	25 bytes/s
300 Bd	300 bits/s	3.333 ms	37.5 bytes/s
600 Bd	600 bits/s	1.667 ms	75 bytes/s
1200 Bd	1200 bits/s	833.333 µs	150 bytes/s
1800 Bd	1800 bits/s	555.556 µs	225 bytes/s
2400 Bd	2400 bits/s	416.667 µs	300 bytes/s
4800 Bd	4800 bits/s	208.333 µs	600 bytes/s
9600 Bd	**9600 bits/s**	**104.167 µs**	**1200 bytes/s**
19200 Bd	19200 bits/s	52.083 µs	2400 bytes/s
28800 Bd	28800 bits/s	34.722 µs	3600 bytes/s
38400 Bd	38400 bits/s	26.042 µs	4800 bytes/s
57600 Bd	57600 bits/s	17.361 µs	7200 bytes/s
76800 Bd	76800 bits/s	13.021 µs	9600 bytes/s
115200 Bd	**115200 bits/s**	**8.681 µs**	**14400 bytes/s**
230400 Bd	230400 bits/s	4.340 µs	28800 bytes/s
460800 Bd	460800 bits/s	2.170 µs	57600 bytes/s
576000 Bd	576000 bits/s	1.736 µs	72000 bytes/s
921600 Bd	921600 bits/s	1.085 µs	115200 bytes/s

e.g. 115200 8N1

115200/(8+2) = 11,520 bytes/s

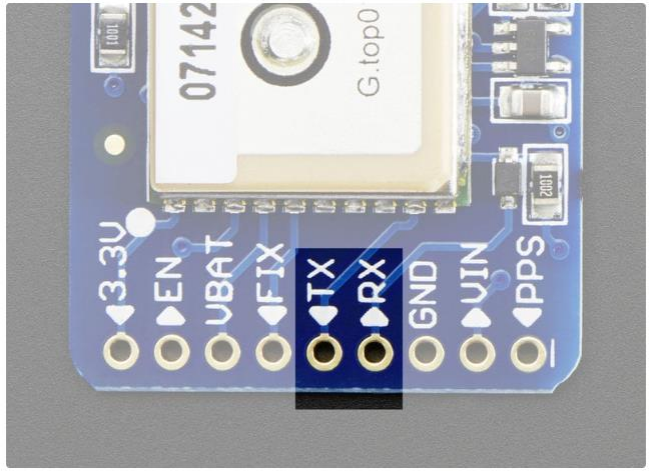
1/11,520 = 86.8 μs per byte

Examples

Adafruit UltimateGPS

https://learn.adafruit.com/adafruit-ultimate-gps/pinouts

Breakout Serial Data Pins



Next pins you'll want to use are the serial data pins:

- **TX** - the pin that transmits data *from* the GPS module to your microcontroller or computer. It's 3.3V logic level. Data comes out at 9600 baud by default
- **RX** - the pin that you can use to send data *to* the GPS. You can use use 3.3V or 5V logic, there is a logic level shifter. By default it expects 9600 baud data, and remember you need to send it checksum'med NMEA sentences

Baud=9600

V_{ref}=3.3V

GasCardNG

(example in tutorial 1 on Blackboard)

1. Procedures for RS232 communication with the Gascard NG

Connect the serial communications port of the Gascard NG to an available communications port on your computer using a suitable cable (see Appendix 1 and Application Note AN034-1for information on how to do this.) and power the Gascard NG. Start the terminal programme you intend to use to communicate with the Gascard NG (e.g. HyperTerminal) and ensure that it is set to the COM port connected to the Gascard NG. (e.g. COM1) set the properties of the port to the following values:

Bits per second	57600
Data bits	8
Parity	None
Stop bits	1
Flow Control	none

57600-8N1

Other protocols

	UART	I2C	SPI
Number of wires	2	2	>3
	115,200	400,000	~2E6
Type	Asynchronous	Synchronous (clock signal used)	Synchronous (clock signal used)
Bytes per frame	1	1-256	1-Unlimited
Bit overhead	2-3	12-270	0
Duplex	Full duplex	Half-duplex	Full duplex
Overhead %	20% - 27%	40% - 11.6%	0%

Further reading
https://www.circuitbasics.com/basics-uart-communication/
https://www.circuitbasics.com/basics-of-the-i2c-communication-protocol/
https://www.circuitbasics.com/basics-of-the-spi-communication-protocol/

General ‘useful’ byte rate equation

- To calculate the useful data rate, we include the effects of overheads

$$\text{byte rate} = \frac{\text{Baud}}{8} \times \frac{\text{(data bits per frame)}}{\text{(total bits per frame)}}$$

$$\text{byte rate} = \frac{\text{Baud}}{8} \times \text{(1-overhead)}$$

Serial Packet Frame Rate

- Mid-level of communication
- The serial message frame per second
- Number of packages of data per second
- Serial frames generally have a similar structure
  - Start byte
  - Header
  - Payload
  - Checksum

Frame Rate

- Example 3-axis accelerometer
  - 1 start byte
  - 4 info/routing bytes
  - 14 data bytes (payload)
  - 1 checksum byte

STX	SYSID	SEQ	TO	FROM	a_x	a_x	a_y	a_y	a_z	a_z	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	CHK
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Start byte

Header

Payload
x3 16-bit
x1 64-bit

Checksum

Total bytes per frame = 20
Useful bytes per frame = 14
70% utilisation, 30% overhead

Data Rate

- What if we only want a single value?
  - e.g. a_z for a vertical position controller

STX	SYSID	SEQ	TO	FROM	a_x	a_x	a_y	a_y	a_z	a_z	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	CHK
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Start byte

Header

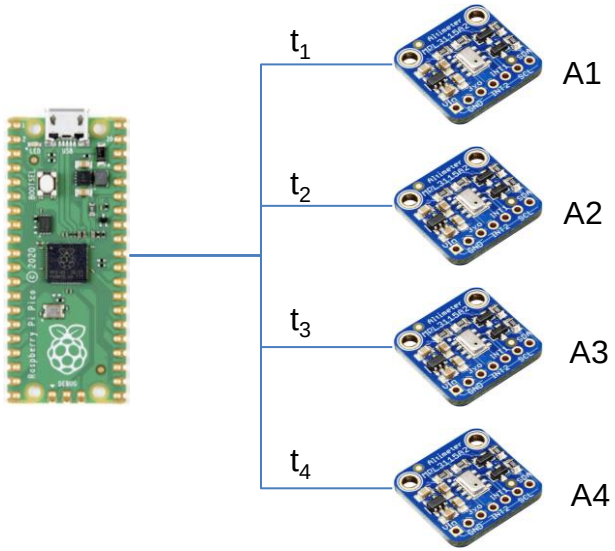
Payload
x3 16-bit
x1 64-bit

Checksum

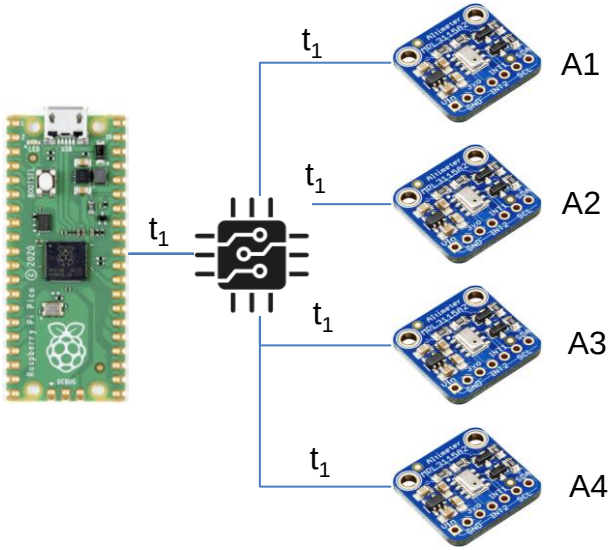
Total bytes per frame = 20
Useful bytes per frame = 2
10% utilisation, 90% overhead!

Multiplexing/addressing

- Use single hardware link for multiple data streams
- Multiplexing – combine data frames into a more complex signal
- Addressing – communicate with individual end point on a time-sharing basis
- For N sensors, and assuming 0% additional overhead, data rate per sensor is reduced by a factor of 1/N



Communicate with each sensor in turn



Additional ‘muxing’ step combines all information into a single larger data frame

Overall ‘useful’ data rate

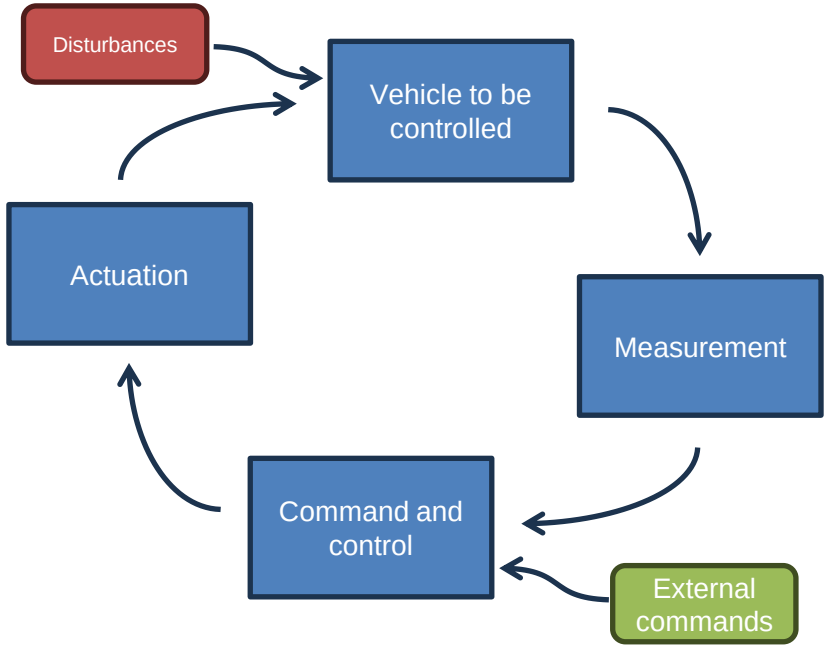
- Combine the bit rates and overheads from the hardware and software stages

$$\text{data rate} = \text{byte rate} \times \frac{\text{(data bytes per frame)}}{\text{(total bytes per frame)}} \times \frac{1}{N}$$

$$\text{data rate} = \frac{\text{Baud}}{8} \times \frac{\text{(data bits per frame)}}{\text{(total bits per frame)}} \times \frac{\text{(data bytes per frame)}}{\text{(total bytes per frame)}} \times \frac{1}{N}$$

Why is data rate important?

- For continuous motion robots, control systems rely on regular updates
- The values in the control loops and algorithms are reliant on sensor measurements at regular intervals
- Generally - cannot control a dynamic behaviour of the vehicle faster than we can sense it



Bandwidth

- Two definitions:

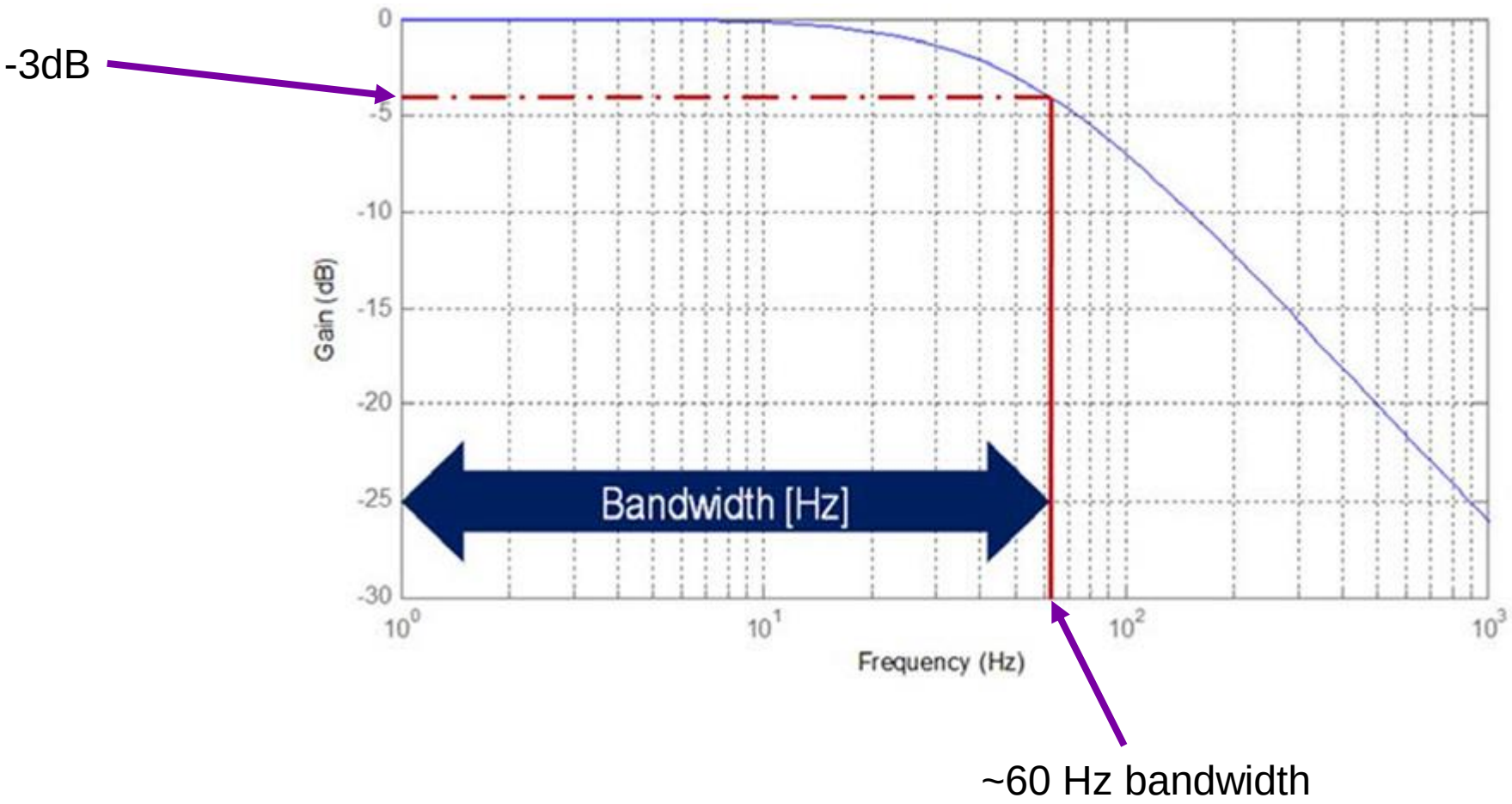
Communications: The maximum possible bit rate of a communication link

Dynamics and Control: “The bandwidth is the first frequency where the gain drops below -3dB of its DC value”

Excitations below this frequency could lead to unstable/growing oscillations. Need to actively sense and control.

Above this frequency energy is lost from the system (70.79% at -3bD). High frequency oscillations are possible but has less effect on overall motion.

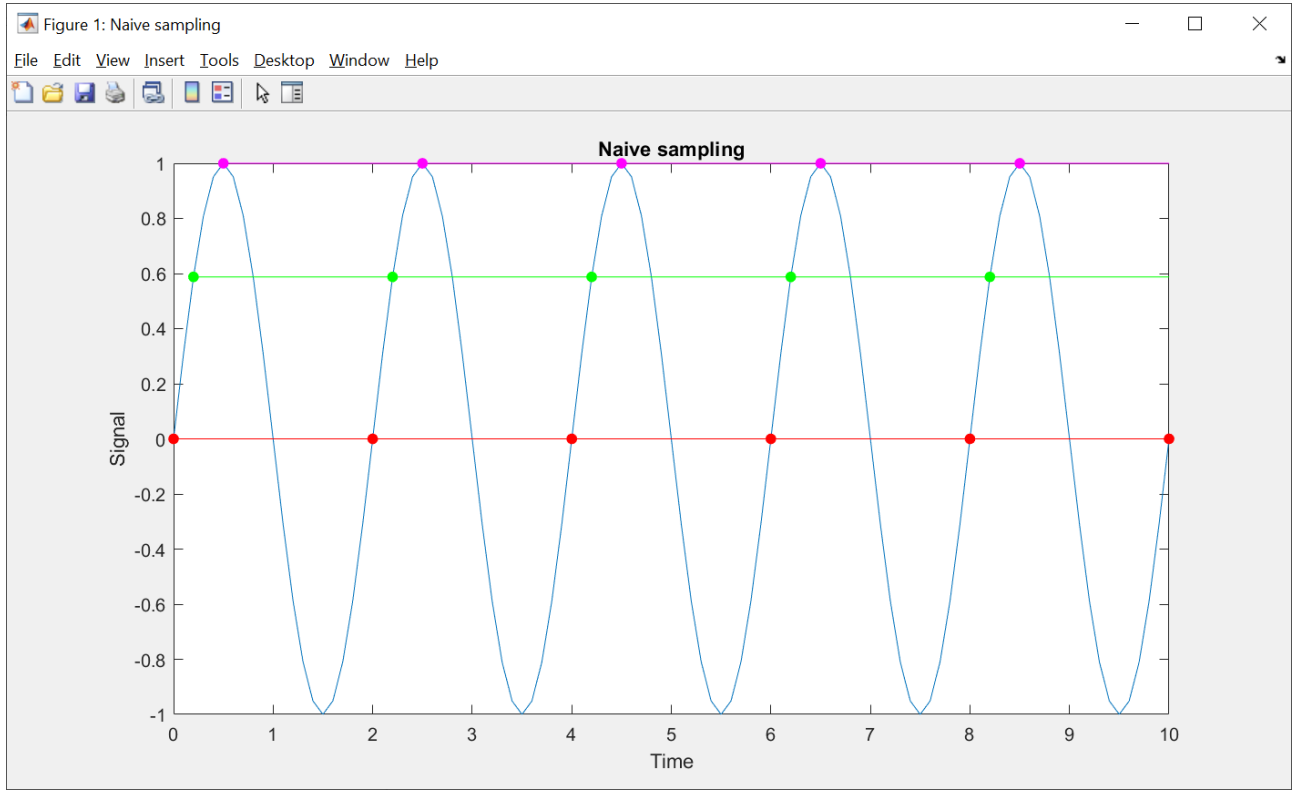
Dynamics bandwidth



Relation between bandwidths

If we have a system with dynamic bandwidth of X Hz, then we must be able to sense and process X Hz signals.

- A **naive** approach would be ensure rates match
- Problem, we cannot represent the full wave form at the bandwidth frequency



Nyquist

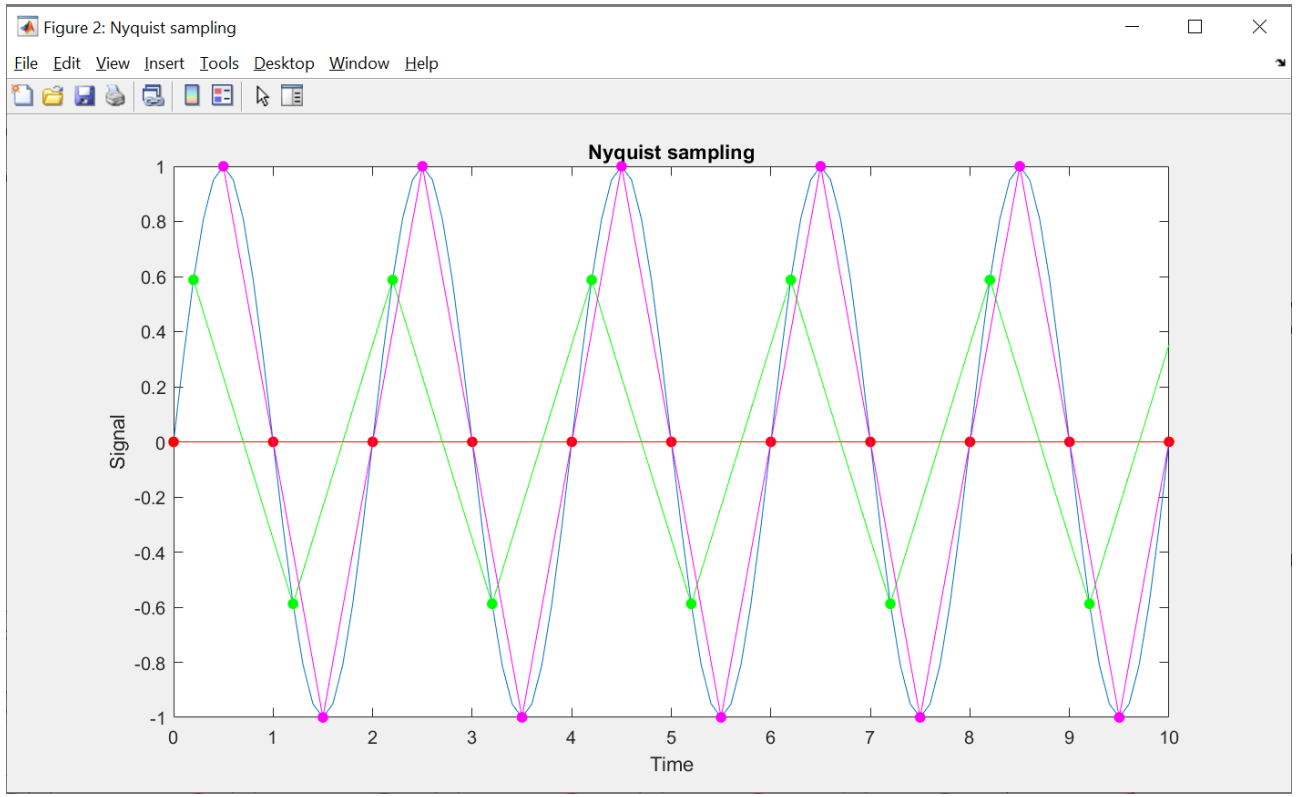
- “The Nyquist theorem specifies that a sinuisoidal function in time or distance can be regenerated with no loss of information as long as it is sampled at a frequency greater than or equal to twice per cycle”
  - Information theory
  - Signal processing
  - Control systems

Pina Colarusso, ... E. Neil Lewis, in Encyclopedia of Spectroscopy and Spectrometry, 1999

Relation between bandwidths

If we have a system with dynamic bandwidth of X Hz, then we must be able to sense and process X Hz signals.

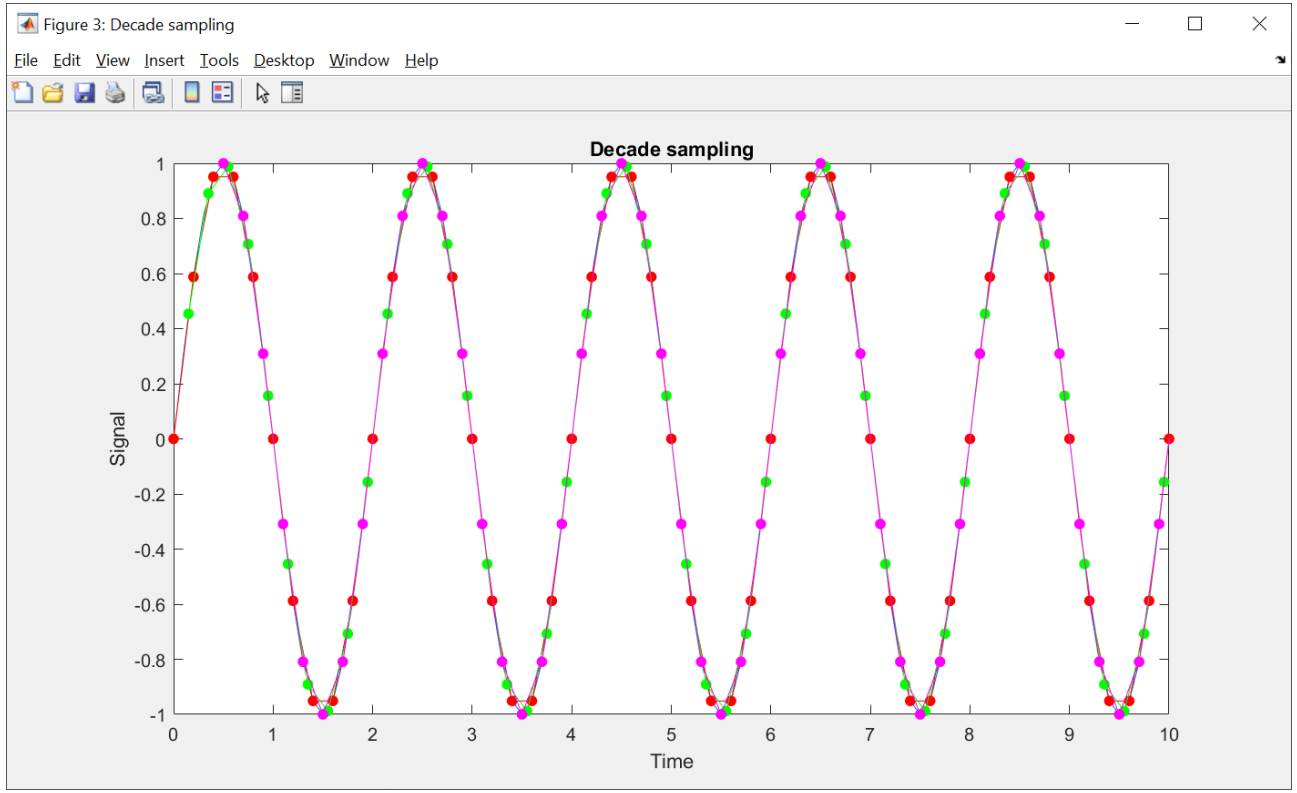
- A **Nyquist** approach at sample rates 2X Hz
- Problem, with real systems we cannot ensure samples will occur at the correct phase



Relation between bandwidths

If we have a system with dynamic bandwidth of X Hz, then we must be able to sense and process X Hz signals.

- A **Power of 10 (decade)** approach at sample rates 10X Hz
- Signal represented with acceptable accuracy at any phase offset



Example

- We have a small agile drone which needs to operate in an indoor environment.
- The electronic speed controllers of the motors accept commands at 50 Hz.
- There is a 0.1 time constant (lag) which affects the propulsion response.
- The IMU (accelerometer and gyroscope) send six 64-bit floating point numbers via I2C communication with a byte (software) overhead of 11.111%, and bit overhead of 12 bits.

- What bit rate will be sufficient for feedback control?

1) the slowest limiting bandwidth of the system is the 0.1 s lag. This has an effective bandwidth of 10 Hz.

2) therefore, we aim for the feedback control loop to operate at a decade higher rate ~100Hz

3) Total bytes per data frame = 64/(1-0.11111111) = 72 bytes

4) Total bits per frame = 72*8 + 12 = 588 bits

5) Combine required rate and bits 588 bits per frame x 100 Hz = 58,800 bit per second

Review

- Leant how to assess the useful data rate
  - Built from hardware bit-rates and software byte rates

- Learn strategies to decide an appropriate sampling rate
  - Based-on bandwidth of dynamic system to be controlled